

Course Code : ECT 355-4

GVHW/RW – 23 / 1219

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

SMART SENSORS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
- (2) Illustrate your answers with a neat diagram wherever necessary.
- (3) All questions are compulsory.

1.
 - (a) Explain the following sensor characteristics :
Accuracy, Reliability, Dead Zone and Non-linearity. 6(CO1)
 - (b) Illustrate resolution of a sensor with an example. How does it differ from accuracy ? 4(CO1)
2.
 - (a) A PT 100 RTD is employed to sense the temperature of a chemical reaction. If the temperature of reaction is 65°C , calculate the resistance of RTD. Assume resistance temperature coefficient of RTD is $0.00392/^{\circ}\text{C}$. 5(CO2)
 - (b) Explain bulk micromachining technique in detail including steps followed in its processing. 5(CO4)
3.
 - (a) Analyze the fabrication technique that is useful in producing the sensor parts of heights ranging from micrometers to centimeters. 5(CO4)
 - (b) Illustrate with an example, how smart sensors differ from base sensors. 5(CO3)
4.
 - (a) What is purpose of IEEE 1451 standards for Smart Sensors ? Can you write a brief outline on IEEE 1451 standards ? 5(CO3)

GVHW/RW-23 / 1219

Contd.

- (b) What do you understand by TEDS in IEEE 1451 standard ? Explain virtual TEDS. 5(CO2)
5. (a) Plastics are typically insulators. Justify briefly the use of plastic as a sensor material. 5(CO2)
- (b) Discuss Silicon as Sensing Material. What kind of physical effects does it exhibits ? 5(CO5)
6. (a) Briefly explain the difference between Intrinsic and Extrinsic Fiber-Optic Sensors. 5(CO5)
- (b) What properties a Smart Sensor must process if it is to be used in an automated consumer applications such as smart cars or smart homes ? 5(CO5)

